

Introduction :

Credit Risk:

$\text{Credit Risk} = \max\{\text{Actual Loss} - \text{Expected Loss}, 0\}$

Credit risk is thus the risk that the actual loss is larger than the expected loss

Expected Loss = Probability of Default × Exposure at Default × Loss Given Default

where:

- *Probability of Default (PD)*: expected probability that a borrower will default on the debt before its maturity, historical default record of other loans with similar characteristics, default probability of a borrower over a one year period. Often given as a rating
- *Exposure at Default (EAD)*: the exact amount the borrower owes at the time of default and can easily be estimated at any time as the current exposure
- *Loss Given Default (LGD)*: percentage of the actual loss of EAD, Banks protect themselves by taking collateral or by holding credit derivatives as a securitization

Credit Modeling:

- *credit rating models*: model PDs
- *credit pricing models*: combinations of PDs, EADs and LGDs to model the EL

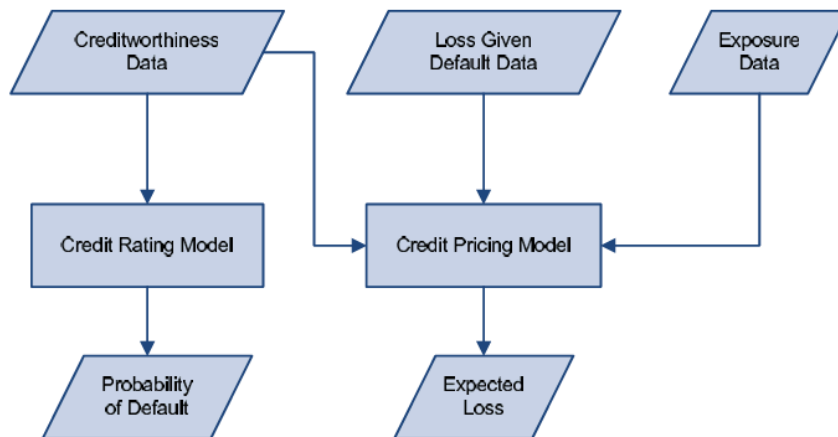


Figure 2.1: Systematic overview of Credit Assessment Models.

The rectangular boxes represent processes, the boxes with the sloped sides represent numerical informations.

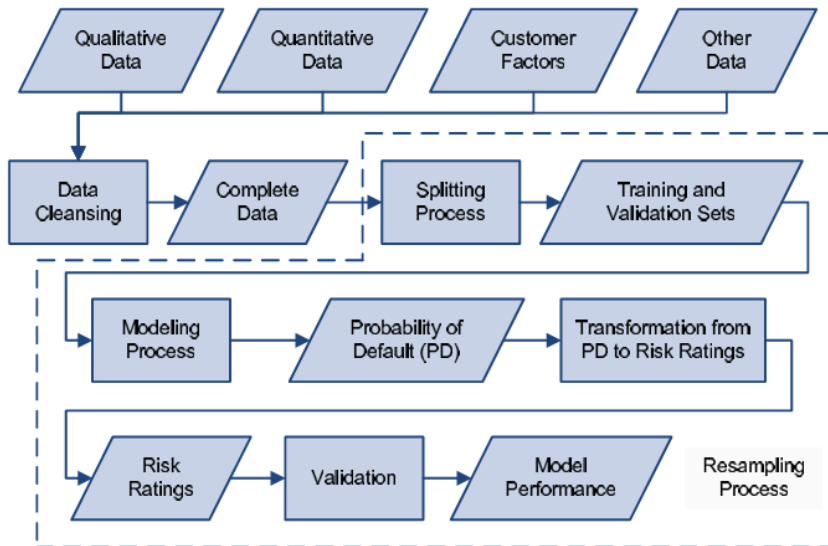


Figure 2.3: Systematic overview of the Credit Rating Modeling Process.

Credit Assessment Models:

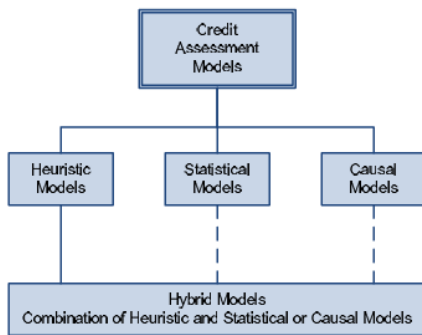


Figure 3.1: Systematic overview of Credit Assessment Models.

Heuristic Models:

- **Rating Questionnaires:** Points based. Frequently observed in the public sector. Example of questions for a public sector customer might be, sex, age, marital status and income.
- **Qualitative Systems:** Grade based. evaluate the applicant for each factor, final assessment would be a weighted or simple average of all grades.
- **Expert Systems:** software solutions which aim to recreate human problem solving abilities
- **Fuzzy Logic Systems:** 0-1

Statistical Models :

- **Discriminant Analysis:** Altman Z-Score

$$Z = 0.12X_1 + 0.14X_2 + 0.033X_3 + 0.006X_4 + 0.999X_5$$

where:

$$X_1 = \text{WorkingCapital}/\text{TotalAssets}.$$

Measures net liquid assets in relation to the size of the company.

$$X_2 = \text{RetainedEarnings}/\text{TotalAssets}.$$

Measures profitability that reflects the company's age

$$X3 = EBIT / TotalAssets.$$

Measures operating efficiency apart from tax and leveraging factors.

$$X4 = MarketValueEquity / BookValueofTotalDebt.$$

Measures how much firms market value can decline before coming insolvent.

$$X5 = Sales / TotalAssets.$$

Standard measure for turnover and varies greatly from industry to industry.

$2.99 < Z\text{-score}$	Firms having <i>low</i> probability of default
$1.81 \leq Z\text{-score} \leq 2.99$	Firms having <i>intermediate</i> probability of default
$1.81 > Z\text{-score}$	Firms having <i>high</i> probability of default

The main advantage of DA, compared to other classification procedures is that the individual weights show the contribution of each explanatory variable. The result of the linear function is then also easy to interpret, as low Z-score is observed it represents a poor loan applicant.

The downside to DA is that it requires the explanatory variables to be normally distributed. Another prerequisite is that the explanatory variables are required to have the same variance for the groups to be discriminated. In practice this is however often thought to be less significant and thus often disregarded.

- **Logistic Regression:**

$$p(X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k)}}$$

- **Classification and Regression Trees (CART):**
- **k-Nearest Neighbor (kNN):**
- ***Support Vector Machine (SVM):
- **Neural Networks:**

Causal Models :

- **Option Pricing Models**
- **Cash Flow Models**
- **Fixed Income Portfolio Analysis**

Hybrid Form Models

Data and Treatment :

Data Dimensions:

- **Quantitative**
- **Qualitative**
- **Customer factors**
- **Other factors and figures**

Data is cleaned by removing newer and retiring companies (not observed in 2 successive years)

Then filtered by removing some of the variables with many missing values than others

Then split into 50% training set, 25% validation set, 25% test set.

To solve the rarity of default cases problem, The average performance of recursive evaluations is used.

Quantitative:

- Liquidity ratios measure the firm's, availability of cash to pay debt.
- Leverage ratios measure the firm's ability to repay long-term debt.

- Profitability ratios measure the firm's use of its assets and control of its expenses to generate an acceptable rate of return.
- Activity ratios measure how quickly a firm converts non-cash assets to cash assets.
- Market ratios measure investor response to owning a company's stock and also the cost of issuing stock. (not taken into consideration)

financial ratios:

- **liquidity**

$$liquidity = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

if liabilities = 0, liquidity = 1000

- **debt/leverage ratio**

$$\frac{\text{Debt}}{\text{EBITDA}} = \frac{\text{Net interest bearing debt}}{\text{Operating profit/loss} + \text{Depreciation/Amortization}}$$

with:

Net interest bearing debt = Subordinary loan capital+long term liabilities +Current liabilities to mortgagebanks+ Current bank liabilities +Current liabilities to group + Current liabilities to owner, etc. -Liquid funds-Securities - Group debt -Outstanding accounts from owner, etc.

if debt < 0, ratio = -1000

if ebitda =< 0, ratio = 1000

- **Return On total Assets**

$$ROA = \frac{\text{Operating profit/loss}}{\frac{1}{2}(\text{Balance Sheet}_0 + \text{Balance Sheet}_{-1})}$$

if only the current balance sheet is available that value is used instead of the average.

Firms that have undergone large investments will generally have lower ROA, Startups have no balance sheet thus given -100

- **Solvency/equity ratio**

$$solvency = \frac{\text{Shareholders' equity}}{\text{Balance sheet}}$$

if b.sheet = 0, solvency = -100

notes:

1. this research treats startups lack of historical financial statements by "Forcing" the startups into the existing model. (solvency/roa = -100)
2. model is built on book value derived financial ratios (outdated/rigid + don't represent market value), therefore predictive power is weakend
3. sector-relative financial ratios model outperforms the simple raw, firm-specific ratios model. Therefore ratios are scaled into key figures called "Scores" noted " $\bar{\alpha}$ " from 1 to 7 (1 being bad, 7 being good). If ratio is nonsense it's given either 1 or 7 (1 indicates poor creditworthiness, 7 indicates positive creditworthiness.)

The final final scores are already adjusted to account for sector-specific norms, it is no longer necessary to analyze the data

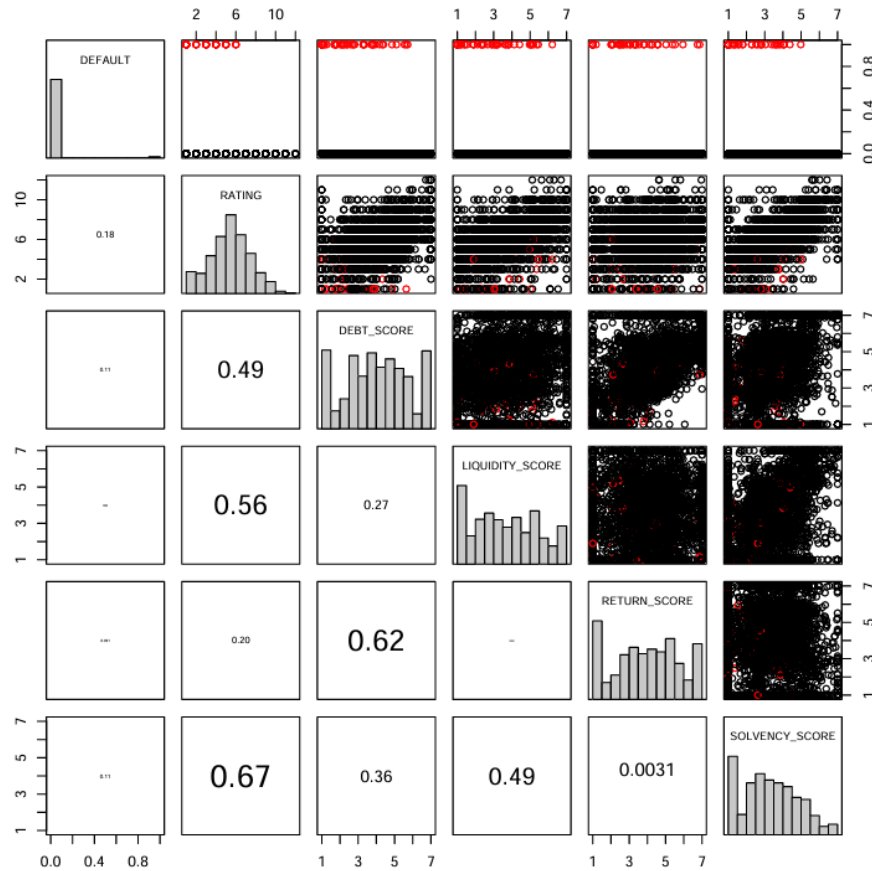


Figure 4.5: Dotplot for all the scaled quantitative factors along with the default variable and RMC ratings, where red dots are defaulted firm and black dots are solvent firms. In the lower triangular the correlation of the variables can be seen and on the diagonal there respective histograms.

separately by sector.

Qualitative:

- Management and strategy
- Sector stability and prospects
- Market position
- Staff situation production facilities and asset assessment
- Financial risk and management
- Refunding

Qualitative ratings from 1 to 7 (1 being bad, 7 being good) handled by the customer chief responsible for the loan application
If values are missing, we use PCA (principal component analysis method)

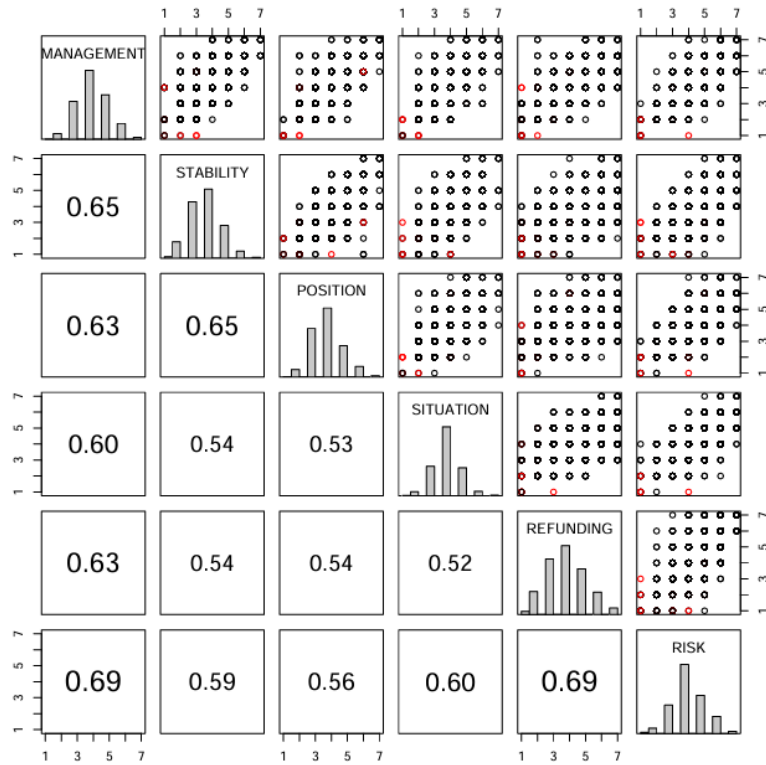


Figure 4.6: Dotplot for all the qualitative factors, where red dots are defaulted firm and black dots are solvent firms. In the lower triangular there is the correlation of the qualitative factors and on the diagonal there are histograms of them.

Qualitative figures are referred to as " ϕ "

Customer factors:

Noted " γ "

Customer Factor	Factor level
Is there an accountants's annotation in the financial statements?	Yes, an outright reservation
	Yes, a supplementary remark
	No
Has the company failed to perform its obligation to FIH?	Yes, within the past year
	Yes, within the past 12-24 months
	No
Is the company listed on the stock exchange?	No
	Yes, but the shares are not OMXC20-listed
	Yes, and the shares are OMXC20-listed
Age of company with current operation?	Up to and including 24 months
	From 25 months and up to and including 60 months
	From 61 months and older

Table 4.7: Customer factors used in the Corporate Model.

Other factors and figures:

- **RMC:**

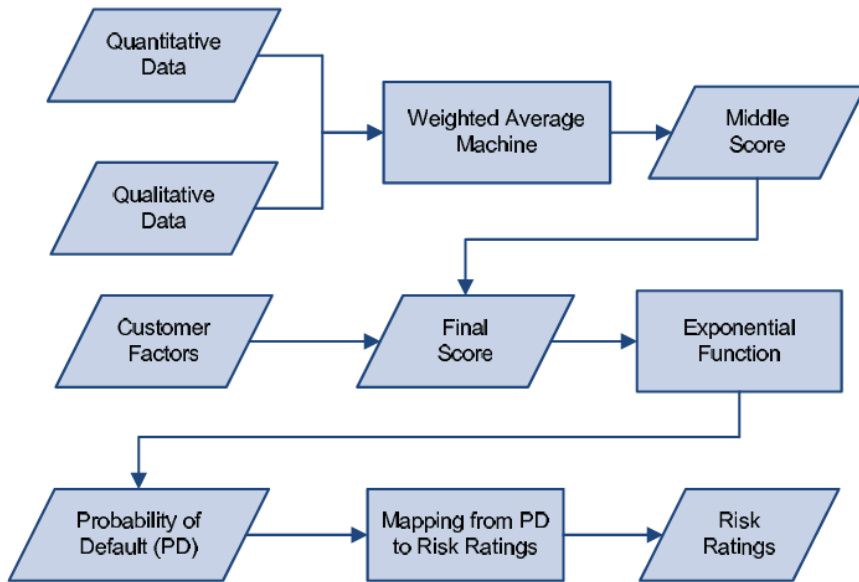


Figure 4.7: Systematic overview of Rating Model Corporate.

PDs are mapped to a final score which is on the range 1- 12.

- **KOB:**

KOB Credit Rating	Risk
B	Very High/Unknown
0-14	Very high
15-33	High
34 - 49	Moderate
50 - 69	Normal
70 - 80	Low
81 -100	Very low

Table 4.8: Creditworthiness of credit rating of the KOB model.

Factors in the KOB Model	Weight
Master data	25%
- Buisness sector	
- Age	
- Numer of Employes	
Economical data	50%
- Solvency	
- Return on Equity	
- Liquidity	
- Netto results	
- Equity	
Other data	25%
- Payment History	
- Accountant's Annotation	
- Qulitative measure	

Table 4.9: Creditworthiness of credit rating of the KOB model.

If score $\Rightarrow$ B50, score = 20. else if score < B50, score = 10

- **Other factors:**
 - Lowest/Highest Earlier Rating (last 12 months)
 - Guarantor Rating (firm that adopts debt in case of default)
 - Subjective Rating (credit expert opinion if some external factors influences creditworthiness)
 - Firms Identity Number
 - Default
 - Equity

Treatment:

General Linear Models:

Linear Regression:

Least Squares Estimation:

Hypothesis Testing:

Goodness of Fit:

Normality:

Homoscedasticity:

Linearity:

Generalized Linear Models:

Exponential family of Distributions:

Logistic Regression:

Model Selection and Assessment:

Discriminant Analysis:

Linear Discriminant Analysis:

Quadratic Discriminant Analysis:

Support Vector Machine:

k-Nearest Neighbors:

CART:

Principal Component Analysis

Solving PCA, eigenvalue decomposition:

Validation :

Relative and Cumulative frequencies:

ROC Curves:

Measuring Discriminatory Power:

AUC:

Gini Coefficient:

Pietra Index:

Conditional Information Entropy Ratio:

Brier Score:

$$BS = \text{Variance} + \text{Calibration} - \text{Resolution}$$

$$BS = p(1 - p) + \frac{1}{N} \sum_{c=1}^K N_c (\hat{p}_c - p_c)^2 - \frac{1}{N} \sum_{c=1}^K N_c (p - p_c)^2$$

Reliability Diagrams:

Stability Analysis:

Results:

General Results:

- **Discriminatory Power:** The LR model achieved a higher Area Under the ROC Curve (AUC) and superior statistical separation scores, proving it is significantly better at distinguishing between solvent and defaulting firms.
- **Rating Distribution:** While the RMC heavily clustered firms in the middle rating tiers, the LR model utilized the entire 1–12 rating scale more effectively, establishing a clearer gradient between credit profiles.
- **Calibration Accuracy:** Reliability diagrams revealed that the RMC was systematically miscalibrated and overly pessimistic, forecasting default rates much higher than observed reality. The LR model's probability predictions tracked much closer to the optimal baseline.
- **Portfolio Stability:** Transition matrices demonstrated that the LR model possessed higher year-over-year rating stability (71% vs. 65%), meaning highly-rated corporate clients were far less likely to experience erratic, sudden downgrades in subsequent years.

PCA: